midterm_review - PDF-XChange Editor
Home View Comment Protect Form Organize Convert Share Review Accessibility Bookmarks Help Format
06_Classification_II_notes midterm_review

USC Viterbi
School of Engineering
Daniel J. Epstein Department of Industrial and Systems Engineering

ISE529 Predictive Analytics
Instructor: Dr. Tao Ma
2025 Fall

Midterm Exam Review

1. Consider the following inverse of the model matrix:

$$(X'X)^{-1} = \begin{bmatrix} 0.893758 & -0.028245 & -0.0175641 \\ -0.028245 & 0.0013329 & 0.0001547 \\ -0.017564 & 0.0001547 & 0.0009108 \end{bmatrix}$$

(a) How many variables are in the regression model?
(b) If the estimate of σ^2 is 50, what is the estimate of the variance of each regression coefficient?
(c) What is the standard error of the intercept?

2. You have fit a regression model with two regressors to a data set that has 20 observations. The total sum of squares is 1000 and the model sum of squares is 750.
(a) What is the value of R^2 for this model?
(b) What is the adjusted R^2 for this model?
(c) What is the value of the F-statistic for testing the significance of regression? What conclusions would you draw about this model if $\alpha = 0.05$? What if $\alpha = 0.01$?
(d) Suppose that you add a third regressor to the model and as a result, the model sum of squares is now 785. Does it seem to you that adding this factor has improved the model?

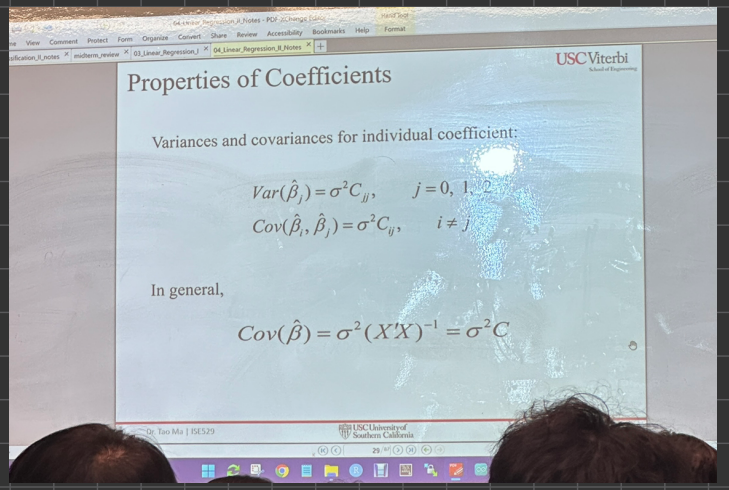
1.

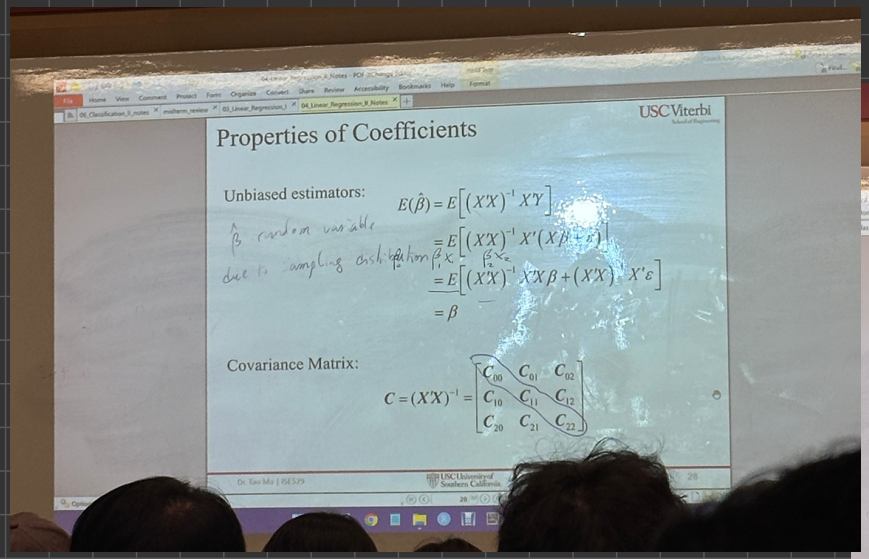
(a) z (one is intercept)

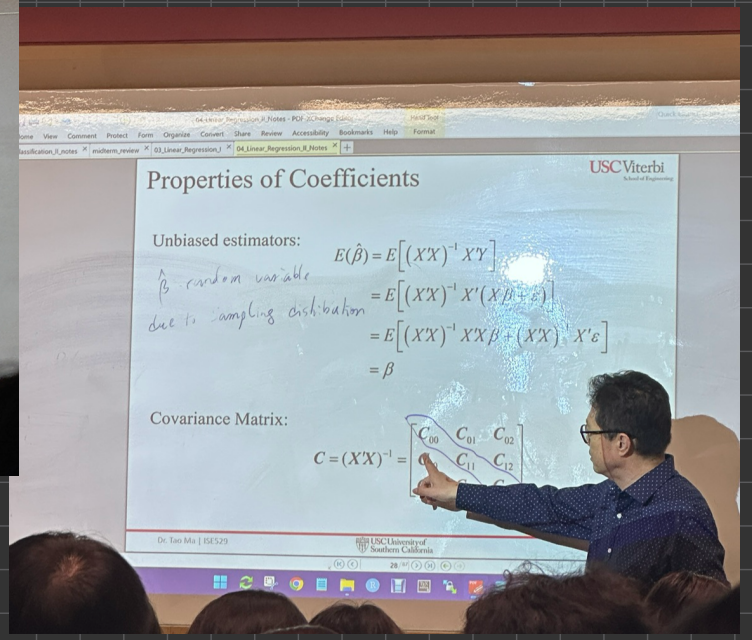
(b) $Var(\beta_1) = \sigma^2 C_{22} = 50 \times 0.0013329 = 0.066645$

$Var(\beta_2) = \sigma^2 C_{33} = 50 \times 0.000908 = 0.04554$

(c) $\sqrt{Var(\beta_0)} = \sqrt{\sigma^2 C_{11}} = \sqrt{50 \times 0.893758} = 6.6849$







2. You have fit a regression model with two regressors to a data set that has 20 observations. The total sum of squares is 1000 and the model sum of squares is 750.
(a) What is the value of R^2 for this model?
(b) What is the adjusted R^2 for this model?
(c) What is the value of the F-statistic for testing the significance of regression? What conclusions would you draw about this model if $\alpha = 0.05$? What if $\alpha = 0.01$?
(d) Suppose that you add a third regressor to the model and as a result, the model sum of squares is now 785. Does it seem to you that adding this factor has improved the model?

3. Answer the following questions:

(a) If the Bayes decision boundary is linear, do we expect LDA or QDA to perform better on the training set? On the test set?

(b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to perform better on the training set? On the test set?

(c) In general, as the sample size n increases, do we expect the test prediction accuracy of QDA relative to LDA to improve, decline, or be unchanged? Why?

(d) True or False: Even if the Bayes decision boundary for a given problem is linear, we will probably achieve a superior test error rate using QDA rather than LDA because QDA is flexible enough to model a linear decision boundary. Justify your answer.

2. $n=20$

$SS_T = 1000$ $\longrightarrow$ $SS_E = SS_T - SS_R$

$SS_R = 750$

(a) $R^2 = \frac{SS_R}{SS_T} = \frac{750}{1000} = 0.75$

(b) $R_{adj}^2 = \frac{SS_E/n-p}{SS_T/n-1} = \frac{(1000-750)/(20-3)}{1000/(20-1)} = 0.7206.$

(c) $F = \frac{MS_R}{MS_E} = \frac{SS_R/k}{SS_E/(n-p)} = \frac{750/2}{250/(20-3)} = 25.5$

$\alpha = 0.05 \Rightarrow f_{0.05,2,17} = 3.59 < F_0$ Reject H_0 $\Rightarrow$ meaningful.

$\alpha = 0.01 \Rightarrow f_{0.01,2,17} = 6.11 < F_0$ Reject H_0

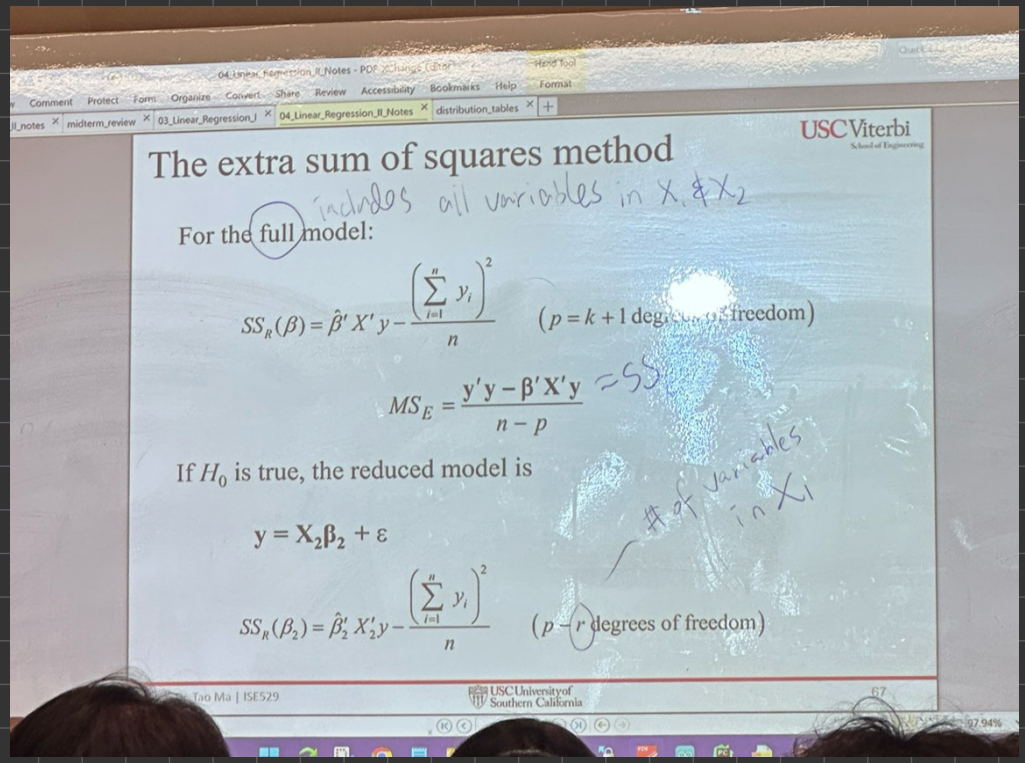
(d) 新增 variable $\Rightarrow$ SST不變.

$SS_R = 750 \rightarrow SS_R' = 785$

$$F' = \frac{(SS_{R\,full} - SS_{R\text{-}reduced})/r}{MS_{E\text{-}full}} = \frac{(785-750)/1}{750/(20-4)} = 2.160$$

r: 相差的 variable 个数量

$f_{0.05,1,16} = 4.49 > F'$



partial-F test ppt

2. You have fit a regression model with two regressors to a data set that has 20 observations. The total sum of squares is 1000 and the model sum of squares is 750.
(a) What is the value of R^2 for this model?
(b) What is the adjusted R^2 for this model?
(c) What is the value of the F-statistic for testing the significance of regression? What conclusions would you draw about this model if $\alpha = 0.05$? What if $\alpha = 0.01$?
(d) Suppose that you add a third regressor to the model and as a result, the model sum of squares is now 785. Does it seem to you that adding this factor has improved the model?

3. Answer the following questions:

(a) If the Bayes decision boundary is linear, do we expect LDA or QDA to perform better on the training set? On the test set?

(b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to perform better on the training set? On the test set?

(c) In general, as the sample size n increases, do we expect the test prediction accuracy of QDA relative to LDA to improve, decline, or be unchanged? Why?

(d) True or False: Even if the Bayes decision boundary for a given problem is linear, we will probably achieve a superior test error rate using QDA rather than LDA because QDA is flexible enough to model a linear decision boundary. Justify your answer.

3.

(a) train → QDA : flexible! lower bias

test → LDA : accurate (True boundary = linear bias should be low)

(b) both QDA = flexible ✓ (non-linear)

(c) improve

(d) false, test error rate : LDA better

QDA → overfitting!

bias-variance trade off

4. There is a set of data ($n = 100$ observations) containing a single predictor and a quantitative response. Then fit a linear regression model to the data, as well as a separate cubic regression, i.e. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$.

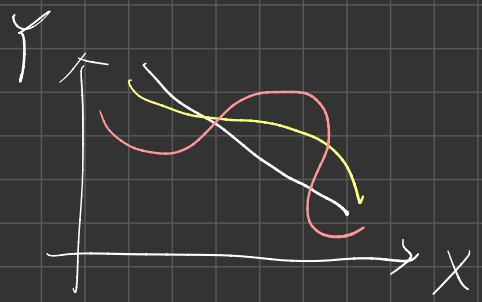
(a) Suppose that the true relationship between X and Y is linear, i.e., $Y = \beta_0 + \beta_1 X + \varepsilon$. Consider the training residual sum of squares (RSS) for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer. How about using test rather than training RSS?

(b) Suppose that the true relationship between X and Y is not linear, but we don't know how far it is from linear. Consider the training RSS for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer. How about using test rather than training RSS?

4. (a) linear would be lower (linear relationship)

(b) no enough info

(not sure ~ or ~)



the cubic regression. Would we expect one to be lower than the other, be the same, or is there not enough information to tell? Justify your answer rather than training RSS?

5. Suppose we collect data for a group of students in a statistics class with variables X_1 = hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.

(a) Estimate the probability that a student who studies for 40 hours and has an undergrad GPA of 3.5 gets an A in the class.

(b) How many hours would the student in part (a) need to study to have a 50 % chance of getting an A in the class?

6. Suppose that we wish to predict whether a given stock will

(5)

$$P(Y=A|X) = \frac{e^{\beta_0+\beta_1X_1+\beta_2X_2}}{1+e^{\beta_0+\beta_1X_1+\beta_2X_2}} \quad \text{if } e^{-6+0.05X_1+X_2}$$

$$P(Y=N|X) = 1 - P(Y=A|X)$$

(a) $X_1 = 40 \quad X_2 = 3.5$

$$P = \frac{e^{-6+0.05\times40+3.5}}{1+e^{\cdots}} = 0.38$$

(b) $P(Y=A|X) = 0.5$

Given $X_2 = 3.5$

$$0.5 = \frac{e^{-6+0.05X_1+3.5}}{1+e^{-6+0.05X+3.5}} \qquad X_1 =$$

6. Suppose that we wish to predict whether a given stock will issue a dividend this year ("Yes" or "No") based on X, last year's percent profit. We examine many companies and discover that the mean value of X for companies that issued a dividend was $\bar{X} = 10$, while the mean for those that didn't was $\bar{X} = 0$. In addition, the variance of X for these two sets of companies was $\hat{\sigma}^2 = 36$. Finally, 80% of companies issued dividends. If X follows a normal distribution, predict the probability that a company will issue a dividend this year given that its percentage profit was $X = 4$ last year.

6.

LDA

$$P(Y=yes \mid X) = \frac{f_k(X) \quad \pi_1}{P(X \mid Y=yes) \cdot P(Y=yes) + P(X \mid Y=no) \cdot P(Y=NO)}$$

$$= \frac{\pi_1 f_1(X)}{\pi_1 f_1(X) + \pi_2 f_2(X)} = \frac{0.8 f_1(X)}{0.8 f_1(X) + 0.2 f_2(X)}$$

$$f_1(X) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(X-\bar{X})^2}{2\sigma^2}}$$

$$f_2(X) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(X-\bar{X})^2}{2\sigma^2}}$$

$$= \frac{0.8 \cdot e^{\frac{-(X-10)^2}{2 \cdot 36}}}{0.8\, e^{\frac{-(X-10)^2}{2 \cdot 36}} + 0.2\, e^{\frac{-(X-0)^2}{2 \cdot 36}}} = 0.752$$

ISE529 Midterm Review 2025 Fall

7. Usually, for two classes coded as (0, 1), the probability from a logistic model can be written as

$$Pr(y=1|X)=\frac{e^{X'\beta}}{1+e^{X'\beta}} \text{ and } Pr(y=0|X)=\frac{1}{1+e^{X'\beta}}$$

However, if two classes are coded as (-1, 1), the probability can be written as

$$Pr(y=1 \text{ or } -1|X)=\frac{1}{1+e^{-y(X'\beta)}}$$

Explain why? Write out Bernoulli PMF function for y = (-1 or 1).

7.

$$Y_i = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

$$P(Y_i) = p_i^{y_i}(1-p)^{1-y_i}$$

$$Y_i = \begin{cases} 1 & \text{success} \\ -1 & \text{failure} \end{cases}$$

$$P(y_i) = p_i^{\frac{y_i+1}{2}}(1-p_i)^{\frac{1-(y_i+1)}{2}}$$